Systems of Linear Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Setting up the augmented matrix**

Write the augmented matrix for the system $2x + y - z = 3$, $x - 3y + 2z = -1$, $4x + y + z = 5$ (give it as its three rows).

Row 1: $\langle 2, 1, -1, 3 \rangle$. Row 2: $\langle 1, -3, 2, -1 \rangle$. Row 3: $\langle 4, 1, 1, 5 \rangle$ (each row lists the x, y, z coefficients, then the right-hand side).

2 **[Foundational]** • **Target: Setting up the augmented matrix**

Write the augmented matrix for the system $x - y = 4$, $3x + 2y = 7$.

Row 1: $\langle 1, -1, 4 \rangle$. Row 2: $\langle 3, 2, 7 \rangle$.

3 **[Foundational]** • **Target: Elementary row operations**

List the three types of elementary row operations.

1. Swap two rows. 2. Multiply a row by a nonzero scalar. 3. Add a multiple of one row to another row.

4 **[Foundational]** • **Target: Elementary row operations**

Given the augmented matrix with rows $\langle 1, 2, 5 \rangle$, $\langle 3, -1, 1 \rangle$, apply the row operation $R_2 \to R_2 - 3R_1$ and give the new Row 2.

$R_2 - 3R_1 = \langle 3, -1, 1 \rangle - 3\langle 1, 2, 5 \rangle = \langle 3 - 3, -1 - 6, 1 - 15 \rangle = \langle 0, -7, -14 \rangle$.

5 **[Foundational]** • **Target: Solving 2x2 systems by elimination**

Solve $2x + 3y = 7$, $x - y = 1$.

From the second equation, $x = 1 + y$. Substituting: $2(1 + y) + 3y = 7 \Rightarrow 5y = 5 \Rightarrow y = 1$, so $x = 2$. Solution: $(2, 1)$.

6 **[Foundational]** • **Target: Solving 2x2 systems by elimination**

Solve $3x - 2y = 4$, $5x + y = 11$.

From the second equation, $y = 11 - 5x$. Substituting: $3x - 2(11 - 5x) = 4 \Rightarrow 13x = 26 \Rightarrow x = 2$, so $y = 1$. Solution: $(2, 1)$.

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7 [Foundational] • Target: Solving 2x2 systems by elimination

Solve $4x + y = 10$, $2x - 3y = -8$.

From the first equation, $y = 10 - 4x$. Substituting: $2x - 3(10 - 4x) = -8 \Rightarrow 14x = 22 \Rightarrow x = \frac{11}{7}$. Then $y = 10 - \frac{44}{7} = \frac{26}{7}$. Solution: $\left(\frac{11}{7}, \frac{26}{7}\right)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving 3x3 systems by Gaussian elimination

Solve $x + y + z = 6$, $2y + 5z = -4$, $2x + 5y - z = 27$.

From the first equation, $x = 6 - y - z$. Substituting into the third:
$2(6 - y - z) + 5y - z = 27 \Rightarrow 3y - 3z = 15 \Rightarrow y = 5 + z$. Substituting into the second:
$2(5 + z) + 5z = -4 \Rightarrow 7z = -14 \Rightarrow z = -2$, so $y = 3$ and $x = 6 - 3 - (-2) = 5$. Solution: $(5, 3, -2)$.

9 [Intermediate] • Target: Solving 3x3 systems by Gaussian elimination

Solve $x + 2y + z = 4$, $y - z = 1$, $3x - y + 2z = 1$.

From the second equation, $y = 1 + z$. Substituting into the first:
$x + 2(1 + z) + z = 4 \Rightarrow x + 3z = 2 \Rightarrow x = 2 - 3z$. Substituting into the third:
$3(2 - 3z) - (1 + z) + 2z = 1 \Rightarrow 6 - 9z - 1 - z + 2z = 1 \Rightarrow 5 - 8z = 1 \Rightarrow z = \frac{1}{2}$. Then $y = \frac{3}{2}$,
$x = 2 - \frac{3}{2} = \frac{1}{2}$. Solution: $\left(\frac{1}{2}, \frac{3}{2}, \frac{1}{2}\right)$.

10 [Intermediate] • Target: Row echelon form and back-substitution

A system's augmented matrix has been reduced to row echelon form with rows $\langle 1, 2, -1, 4\rangle$, $\langle 0, 1, 3, -2\rangle$, $\langle 0, 0, 1, 5\rangle$ (variables x, y, z). Use back-substitution to solve.

From row 3: $z = 5$. From row 2: $y + 3(5) = -2 \Rightarrow y = -17$. From row 1:
$x + 2(-17) - 5 = 4 \Rightarrow x - 39 = 4 \Rightarrow x = 43$. Solution: $(43, -17, 5)$.

11 [Intermediate] • Target: Row echelon form and back-substitution

Explain why row echelon form makes a system easy to solve, in one or two sentences.

In row echelon form, the last nonzero row involves only the last variable, so it can be solved immediately; each preceding row then involves only variables already solved for plus one new one, letting the solution unwind one variable at a time — back-substitution.

12 [Intermediate] • Target: Inconsistent and dependent systems

Solve (or classify) the system $x + y = 3$, $2x + 2y = 8$.

The second equation is $2(x + y) = 8$, i.e., $x + y = 4$ — contradicting $x + y = 3$. The system is inconsistent (no solution): the two lines are parallel and distinct.

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13 [Intermediate] • Target: Inconsistent and dependent systems

Solve (or classify) the system $x + y = 3$, $2x + 2y = 6$.

The second equation is exactly $2\times$ the first, so it carries no new information. The system has infinitely many solutions: $y = 3 - x$ for any real x (the two equations describe the same line).

14 [Intermediate] • Target: Inconsistent and dependent systems

For the system $x - 2y = 4$, $-3x + 6y = k$, find the value of k that makes the system have infinitely many solutions (rather than no solution).

The second equation is -3 times the first row's coefficients ($-3(x - 2y) = -3x + 6y$), so it matches when $-3(4) = k$, i.e., $k = -12$. Any other k gives an inconsistent (no-solution) system.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Why row operations preserve the solution set • Cross-Lesson: 1.01 Vectors in n Dimensions

Explain briefly why multiplying a row by a nonzero constant c doesn't change a system's solution set.

Multiplying both sides of an equation by a nonzero c produces an equivalent equation, since the operation is invertible — dividing by c recovers the original equation exactly.

16 [Challenging] • Target: Why row operations preserve the solution set • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Why is the restriction $c \neq 0$ needed for this row operation?

If $c = 0$, the row becomes the zero row, which is not invertible — the original row's information is destroyed, so the resulting system need not have the same solution set as the original.

17 [Challenging] • Target: Why row operations preserve the solution set • Cross-Lesson: 4.01 Determinants

Explain briefly why adding a multiple of one row to another row also preserves the solution set.

This operation is also invertible: subtracting the same multiple of the row back off recovers the original row exactly. Since any solution to the original system automatically satisfies the new (combined) equation too, and the operation can always be undone, the solution set is unchanged.

18 [Challenging] • Target: Systems with a parameter • Cross-Lesson: 1.01 Vectors in n Dimensions

For what value(s) of k does the system $x + ky = 3$, $2x + 4y = 6$ have infinitely many solutions?

The second equation simplifies to $x + 2y = 3$. This matches the first equation exactly when $k = 2$ (giving two copies of the same line). For $k = 2$, infinitely many solutions; for any other value of k, the two lines are not parallel, so the system has a unique solution.

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19 [Challenging] • Target: Systems with a parameter • Cross-Lesson: 3.01 Matrices & Matrix Algebra

For what value of k does the system $2x - y = 5$, $4x + ky = 1$ have NO unique solution (i.e., the coefficient matrix is singular)?

The system fails to have a unique solution exactly when the two equations' coefficients are proportional: $\frac{4}{2} = \frac{k}{-1} \Rightarrow 2 = -k \Rightarrow k = -2$.

20 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 4.01 Determinants

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 L of a 15% solution. Set up the system of equations (let x, y be the liters of each solution used).

Volume: $x + y = 20$. Acid content: $0.10x + 0.30y = 0.15(20) = 3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Augmented Matrices and Row Echelon Form (REF)

Which matrix represents a valid Reduced Row Echelon Form (RREF)?

a. $[[1, 0, 3], [0, 1, -2], [0, 0, 0]]$ ✓

b. $[[1, 2, 3], [0, 1, -2], [0, 0, 0]]$

c. $[[2, 0, 0], [0, 1, 0], [0, 0, 1]]$

d. $[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$

In RREF, each leading entry (pivot) is 1, is the only non-zero entry in its column, and appears to the right of pivots in previous rows, with zero rows at the bottom.

22 [Foundational] • Target: Augmented Matrices and Row Echelon Form (REF)

How many pivots does a 3×5 matrix have at most?

a. 3 ✓

b. 5

c. 2

d. 15

Because each pivot must reside in a distinct row and a distinct column, the maximum number of pivots is $\min(m, n) = \min(3, 5) = 3$.

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23 [Foundational] • Target: Augmented Matrices and Row Echelon Form (REF)

An augmented matrix has a row of the form $[0, 0, \ldots, 0 \mid b]$ with $b \neq 0$. What does this establish about the linear system?

a. The system is inconsistent (has no solution) ✓

b. The system has a unique solution

c. The system has infinitely many solutions

d. The determinant is equal to b

The row represents the contradictory equation $0 = b$. Any system with such a pivot in the augmented column is inconsistent.

24 [Foundational] • Target: Solution Sets and Free Variables

A consistent system has 5 variables and 3 pivot columns in its coefficient matrix. How many free variables does the system have?

a. 2 ✓

b. 3

c. 5

d. 0

The number of free variables equals total variables minus pivot columns: $5 - 3 = 2$ free variables, yielding a 2-dimensional plane of solutions.

25 [Foundational] • Target: Solution Sets and Free Variables

When does a linear system $A\mathbf{x} = \mathbf{b}$ possess a unique solution?

a. When the system is consistent and every column of A contains a pivot (no free variables) ✓

b. Whenever the number of equations equals the number of variables

c. Whenever $\mathbf{b} \neq \mathbf{0}$

d. When there is at least one free variable

A consistent system has a unique solution if and only if every variable is a basic variable, meaning there are zero free variables.

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26 [Foundational] • Target: Solution Sets and Free Variables

If a system is consistent and has at least one free variable, how many solutions exist over $\mathbb{R}$?

a. Infinitely many ✓

b. Exactly one

c. None

d. Finite number equal to the number of free variables

Because free variables can take any real value, any consistent system with at least one free variable has infinitely many solutions.

27 [Foundational] • Target: Homogeneous Systems and Null Space

What is always true about the homogeneous linear system $A\mathbf{x} = \mathbf{0}$?

a. It is always consistent, because $\mathbf{x} = \mathbf{0}$ is always a solution (the trivial solution) ✓

b. It has no solutions if $\det(A) = 0$

c. It always has infinitely many solutions

d. It only has solutions when A is square

Setting all variables to zero ($x_1 = \ldots = x_n = 0$) always satisfies $A\mathbf{0} = \mathbf{0}$, so homogeneous systems are never inconsistent.

28 [Foundational] • Target: Homogeneous Systems and Null Space

Under what condition does the homogeneous system $A\mathbf{x} = \mathbf{0}$ have non-trivial solutions (solutions where $\mathbf{x} \neq \mathbf{0}$)?

a. If and only if the system has at least one free variable ✓

b. If and only if A has more rows than columns

c. If and only if all entries of A are positive

d. Whenever $\mathbf{b} = \mathbf{0}$

Non-trivial solutions exist if and only if there is at least one free variable (i.e. the columns of A are linearly dependent).

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Homogeneous Systems and Null Space**

The set of all solutions to $A\mathbf{x} = \mathbf{0}$ forms a subspace of $\mathbb{R}^n$ called the:

a. **Null space (or kernel) of A** ✓

b. Column space of A

c. Row space of A

d. Span of $\mathbf{b}$

By definition, $\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$ is the null space of the matrix.

30 **[Intermediate]** • **Target: Matrix Invertibility and Systems**

According to the Invertible Matrix Theorem, for a square $n \times n$ matrix A, which statement is equivalent to A being invertible?

a. **The equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in \mathbb{R}^n$** ✓

b. The columns of A are linearly dependent

c. $\det(A) = 0$

d. The null space of A contains non-zero vectors

Invertibility is equivalent to A having n pivots, meaning $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x} = A^{-1}\mathbf{b}$ for any $\mathbf{b}$.

31 **[Intermediate]** • **Target: Matrix Invertibility and Systems**

If A is invertible and $A\mathbf{x} = \mathbf{b}$, how is the unique solution expressed algebraically?

a. $\mathbf{x} = A^{-1}\mathbf{b}$ ✓

b. $\mathbf{x} = \mathbf{b}A^{-1}$

c. $\mathbf{x} = \frac{\mathbf{b}}{A}$

d. $\mathbf{x} = A^T\mathbf{b}$

Multiply both sides on the left by A^{-1}: $A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{b} \Rightarrow I\mathbf{x} = A^{-1}\mathbf{b} \Rightarrow \mathbf{x} = A^{-1}\mathbf{b}$.

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32 [Intermediate] • Target: Matrix Invertibility and Systems

If A is a 4×4 matrix with rank 3, how many solutions does $A\mathbf{x} = \mathbf{0}$ have?

a. Infinitely many ✓

b. Exactly one

c. None

d. 3

By the Rank-Nullity Theorem, $\text{nullity}(A) = n - \text{rank}(A) = 4 - 3 = 1$. With a 1-dimensional null space, there are infinitely many solutions.

33 [Intermediate] • Target: Matrix Invertibility and Systems

Consider the following system of linear equations represented by the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

a. 1 ✓

b. 2

c. 3

d. 4

We are given the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$ To transform this matrix into row-echelon form, we can perform the following row operation: $R_2 \leftarrow R_2 - 2R_1$.
$(1, 2, |, 5; 2 - 2(1), -1 - 2(2), |, 0 - 2(5))$ $(1, 2, |, 5; 0, -5, |, -10)$ Now, we can perform another row operation: $R_2 \leftarrow R_2/-5$. $(1, 2, |, 5; 0, 1, |, 2)$ This matrix is now in row-echelon form. The corresponding system of equations is: $x + 2y = 5$ $y = 2$ Substitut.

34 [Intermediate] • Target: Setting up the augmented matrix

Consider the system of linear equations represented by the following augmented matrix: $(1, 2, |, 5; 0, 1, |, 2)$ Use row reduction to determine the unique solution for x and y in the system. What is the value of x?

a. 1

b. 2

c. 3 ✓

d. 4

The augmented matrix represents the system of equations: 1) $x + 2y = 5$ 2) $y = 2$ Since the second equation directly gives the value of y as 2, we can substitute this into the first equation to find x. Substitute $y = 2$ into equation (1): $x + 2(2) = 5$ $x + 4 = 5$ $x = 5 - 4$ $x = 1$ Therefore, the unique solution is $x = 1$ and $y = 2$. The question asks for the value of x.

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35 [Intermediate] • Target: Elementary row operations

Consider the following system of linear equations represented by an augmented matrix: $[1, 2, |, 5; 0, 1, |, 2]$. What is the value of x in the first equation?

a. 1

b. 2

c. **3** ✓

d. 4

The augmented matrix represents the system of equations: $x + 2y = 5$ and $y = 2$. We are asked to find the value of x. We can use substitution to solve for x. Substitute the value of y from the second equation into the first equation: $x + 2(2) = 5$. This simplifies to $x + 4 = 5$. Subtracting 4 from both sides gives $x = 1$. Therefore, the value of x in the first equation is 1.

36 [Intermediate] • Target: Solving 2x2 systems by elimination

Solve for z: $x + y + z = -8$, $y + z = 8$, $z = -4$ (already row-echelon).

a. -16

b. 12

c. **-4** ✓

d. -2

Back-substitute: $z = -4$; $y = 8 - (-4) = 12$; $x = -8 - (12) - (-4) = -16$.

37 [Intermediate] • Target: Solving 3x3 systems by Gaussian elimination

Solve for z: $x + y + z = -5$, $y + z = -4$, $z = -2$ (already row-echelon).

a. -1

b. 0

c. -4

d. **-2** ✓

Back-substitute: $z = -2$; $y = -4 - (-2) = -2$; $x = -5 - (-2) - (-2) = -1$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Row echelon form and back-substitution • Cross-Lesson: 1.01 Vectors in n Dimensions

Solve for x: $x + y + z = -4$, $y + z = 4$, $z = -7$ (already row-echelon).

a. 11

b. -7

c. **-8** ✓

d. -6

Back-substitute: $z = -7$; $y = 4 - (-7) = 11$; $x = -4 - (11) - (-7) = -8$.

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39 [Challenging] • Target: Inconsistent and dependent systems • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Solve for y: $x + y + z = 8$, $y + z = -7$, $z = -3$ (already row-echelon).

a. **-4** ✓

b. 15

c. -3

d. -2

Back-substitute: $z = -3$; $y = -7 - (-3) = -4$; $x = 8 - (-4) - (-3) = 15$.

40 [Challenging] • Target: Why row operations preserve the solution set • Cross-Lesson: 4.01 Determinants

Solve for y: $x + y + z = 7$, $y + z = -7$, $z = -5$ (already row-echelon).

a. 14

b. -5

c. **-2** ✓

d. 0

Back-substitute: $z = -5$; $y = -7 - (-5) = -2$; $x = 7 - (-2) - (-5) = 14$.

41 [Challenging] • Target: Systems with a parameter • Cross-Lesson: 1.01 Vectors in n Dimensions

Solve for z: $x + y + z = 3$, $y + z = -8$, $z = 4$ (already row-echelon).

a. 11

b. -12

c. **4** ✓

d. 6

Back-substitute: $z = 4$; $y = -8 - (4) = -12$; $x = 3 - (-12) - (4) = 11$.

42 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Solve for z: $x + y + z = 4$, $y + z = -2$, $z = -1$ (already row-echelon).

a. 6

b. 1

c. **-1** ✓

d. 0

Back-substitute: $z = -1$; $y = -2 - (-1) = -1$; $x = 4 - (-1) - (-1) = 6$.

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43 [Challenging] • Target: Augmented Matrices and Row Echelon Form (REF) • Cross-Lesson: 4.01 Determinants

Solve for y: $x + y + z = 2$, $y + z = -8$, $z = 5$ (already row-echelon).

a. 10

b. **-13** ✓

c. 5

d. -11

Back-substitute: $z = 5$; $y = -8 - (5) = -13$; $x = 2 - (-13) - (5) = 10$.

44 [Challenging] • Target: Solution Sets and Free Variables • Cross-Lesson: 1.01 Vectors in n Dimensions

Solve for y: $x + y + z = 6$, $y + z = -4$, $z = 6$ (already row-echelon).

a. 10

b. **-10** ✓

c. 6

d. -8

Back-substitute: $z = 6$; $y = -4 - (6) = -10$; $x = 6 - (-10) - (6) = 10$.

45 [Challenging] • Target: Homogeneous Systems and Null Space • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Solve for z: $x + y + z = -4$, $y + z = -1$, $z = -1$ (already row-echelon).

a. -3

b. **-1** ✓

c. 0

d. 1

Back-substitute: $z = -1$; $y = -1 - (-1) = 0$; $x = -4 - (0) - (-1) = -3$.

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